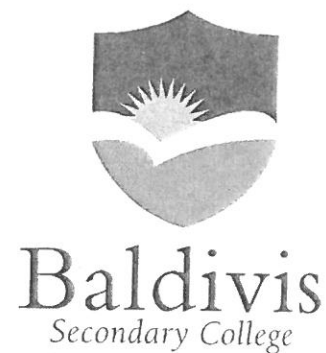


Name: Mah Kinkey

Teacher: Evans

Class: Master



**YEAR 9 EXAM
SEMESTER TWO 2015**

SECTION TWO Resource Allowed

MATERIAL(S) REQUIRED / RECOMMENDED FOR THIS PAPER

TO BE PROVIDED BY THE SUPERVISOR:

This Question / Answer Booklet

TO BE PROVIDED BY THE CANDIDATE:

Standard items: pens, pencils, pencil sharpener, eraser, correction fluid, ruler, highlighters.

Special items: Calculator, 1 x A4 sheet of notes (double sided)

Important note to candidates:

No other items may be used in this section of the examination. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Instructions to candidates

- Candidates are to remain seated during the exam and are not allowed to talk to other candidates.
- Mobile phones are not to be used at all during the exam.
- Students who fail to follow examination rules will be removed and dealt with in accordance to the schools policy.

Write your answers in the spaces provided in this Question/Answer Booklet. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use spare pages for planning, indicate this clearly at the top of the page
- Continuing an answer: If you need the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

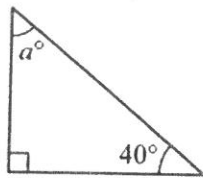
Show all your working clearly. Your working should be sufficient to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat

<i>Question</i>	<i>Out of</i>	<i>Mark</i>
1	4	
2	6	
3	4	
4	4	
5	4	
6	10	
7	5	
8	3	
TOTAL	40	
Your Mark	/40	
Percentage		

1 Find the value of each pronumeral

[4]

a



$$a = 90 - 40$$

$$a = \underline{50^\circ}$$

b



$$180 - 120$$

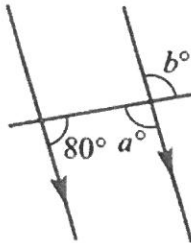
$$2b = 60$$

$$b = \underline{30^\circ}$$

2 Find the value of each of the pronumerals. Give reasons for your answers.

[6]

a

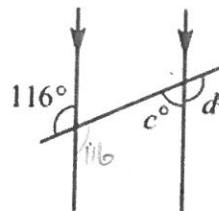


$$180 - 80 = a$$

$$a = 100^\circ \text{ co-interior}$$

$$b = 100^\circ \text{ vertically opposite}$$

b



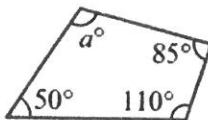
$$c = 64^\circ - \text{co-interior}$$

$$d = 116^\circ - \text{straight line equal } 180^\circ$$

3 Find the value of the pronumeral in each of these polygons.

[4]

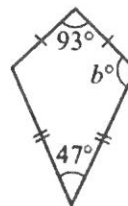
a



$$a = 360 - (50 + 85 + 110)$$

$$a = \underline{115^\circ}$$

b



$$2b = 360 - (93 + 47)$$

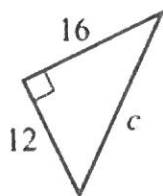
$$2b = 220$$

$$b = \underline{110^\circ}$$

- 4 Find the length of the hypotenuse in each of the following right-angled triangles.
(Round to 2 decimal places where necessary)

[4]

a

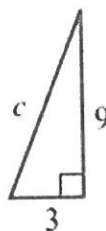


$$c^2 = 16^2 + 12^2$$

$$c = \sqrt{400}$$

$$c = 20$$

b



$$c^2 = 9^2 + 3^2$$

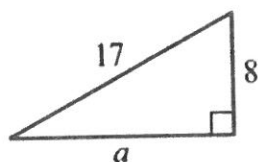
$$c = \sqrt{90}$$

$$c = 9.49$$

- 5 Find the value of the pronumeral in each of the following right-angled triangles
(Round to 2 decimal places where necessary)

[4]

a

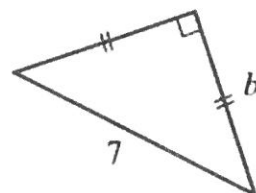


$$a^2 = 17^2 - 8^2$$

$$a = \sqrt{225}$$

$$a = 15$$

b



$$7^2 = b^2 + b^2$$

$$2b^2 = 7^2$$

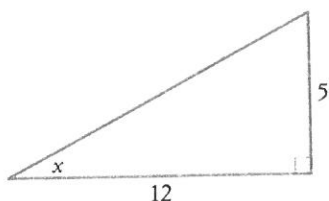
$$b = \sqrt{24.5}$$

$$b = 4.95$$

- 6 Find the values of the pronumerals in each triangle. Express angles and answers to 2 dp.

[10]

a

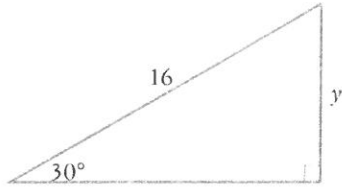


$$\tan x = \frac{5}{12}$$

$$x = \tan^{-1}\left(\frac{5}{12}\right)$$

$$x = 22.62^\circ$$

b

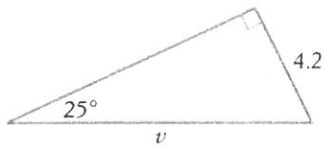


$$\sin 30 = \frac{y}{16}$$

$$y = 16 \times \sin 30$$

$$y = 8$$

c

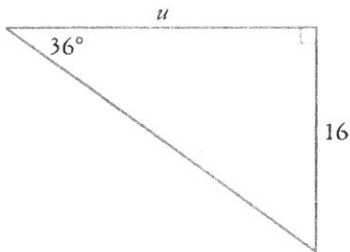


$$\sin 25 = \frac{4.2}{v}$$

$$v = \frac{4.2}{\sin 25}$$

$$v = 9.94$$

d

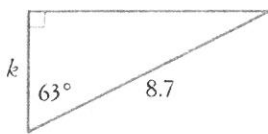


$$\tan 36 = \frac{16}{u}$$

$$u = \frac{16}{\tan 36}$$

$$u = 22.02$$

e

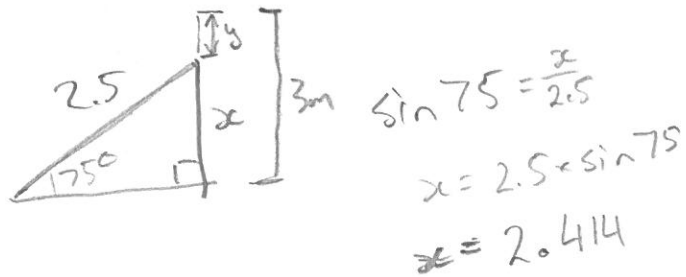
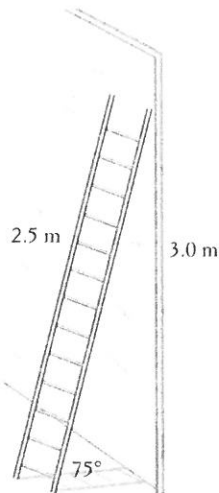


$$\cos 63 = \frac{k}{8.7}$$

$$k = 8.7 \times \cos 63$$

$$k = 3.95$$

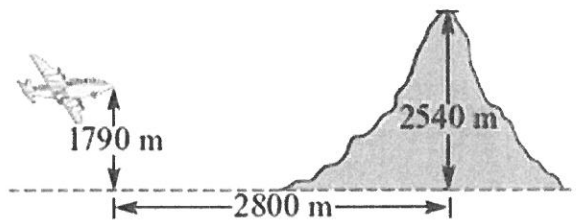
- 7 A 2.5 m long ladder leans against a 3.0 m high wall. If it rests against the (horizontal) ground at an angle of 75° , how far from the top of the wall will it reach? [5]



$$3\text{m} - 2.414 = 0.585$$

$\therefore$ it is 0.585m from the top of the wall

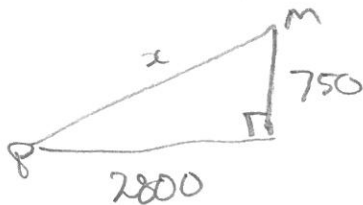
- 8 A pilot flying a plane at an altitude of 1790 m sees a mountain peak 2540 m high, 2800 m away from him in the horizontal direction. [3]



- a What is the difference in the vertical height between the plane and mountain peak?

$$2540 - 1790 = 750\text{m}$$

- b Find the direct distance between the plane and the mountain peak, correct to the nearest metre.



$$x^2 = 2800^2 + 750^2$$

$$x = \sqrt{8402500}$$

$$x = 2898.706\text{m}$$

$\therefore$ direct distance is 2899m